

Basics of Soil Stabilization

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This unit introduces the concept of soil stabilization and its necessity for improving the performance of problematic soils like black cotton soil in infrastructure development. It outlines the fundamental objectives of stabilization, such as improving load-bearing capacity, reducing plasticity, and minimizing shrink-swell behavior. Students will learn about different types of stabilization—mechanical, chemical, and biological—and the factors influencing the choice of stabilization method. Applications in roads, embankments, and foundations are also discussed to emphasize real-world relevance.